

BPHO ROUND 0 · MARKSCHEME

Drill bank section 5

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.*

Quick key

Q1 A	Q2 C	Q3 E	Q4 B	Q5 D
Q6 D	Q7 A	Q8 C	Q9 E	Q10 B
Q11 B	Q12 D	Q13 A	Q14 C	Q15 E
Q16 E	Q17 B	Q18 D	Q19 A	Q20 C
Q21 A	Q22 E	Q23 B	Q24 D	Q25 A

Worked solutions

Q1 · Dimensional analysis as a construction: which combination of a density, a speed and a length has the dimensions of a mass flow rate — **Answer A**

What is being tested. Whether you can *build* a combination rather than recognise one. The five options all look like plausible products of the same three symbols, and reading them off by eye is a guess. Writing one equation for each base dimension turns it into arithmetic.

Step 1 — write the dimensions down. Each quantity is given in terms of mass M, length L and time T:

$$[\rho] = M L^{-3} \quad [v] = L T^{-1} \quad [d] = L$$

target: $M T^{-1}$

The target has no length in it at all. That is the single most useful fact here, and it is what the last equation will have to arrange.

Step 2 — suppose the answer is a product of powers. Any combination of the three must have the form

$$\rho \text{ to the } a \times v \text{ to the } b \times d \text{ to the } c$$

where a , b and c are numbers still to be found. Multiplying powers adds their exponents, so the dimensions of this product are

$$M \text{ to the } a \times L \text{ to the } (-3a + b + c) \times T \text{ to the } (-b)$$

Step 3 — match the mass. The target carries M to the first power and only the density contributes any mass:

$$a = 1$$

Step 4 — match the time. The target carries T to the minus one and only the speed contributes any time:

$$-b = -1, \text{ so } b = 1$$

Step 5 — match the length. The target carries no length at all, so the length exponent must come to zero. This is the only equation that needs real work, because all three quantities bring a length:

$$\begin{aligned} -3a + b + c &= 0 \\ -3(1) + (1) + c &= 0 \\ c &= 2 \end{aligned}$$

Step 6 — assemble it. With $a = 1$, $b = 1$ and $c = 2$,

$$\rho \, v \, d \, \text{squared}$$

So **Answer: A.**

Step 7 — substitute back and check. Put the dimensions in and see whether they really do give the target:

$$M \, L^{-3} \times L \, T^{-1} \times L \, \text{squared} = M \, L \, \text{to the } (-3 + 1 + 2) \, T^{-1} = M \, T^{-1}$$

The lengths cancel exactly, which was the whole requirement. Notice how the answer was forced: two of the three exponents were read straight off the mass and time equations, and only the third needed solving. Three unknowns, three equations, no guessing.

Step 8 — check it makes physical sense. A density times a speed times an area is a mass per unit volume, times a distance per second, times an area. The volume per second is area times speed, and multiplying that by the density gives the mass flowing past each second. So the combination is not just dimensionally correct, it is the right physical statement: mass flow rate = density $\times$ speed $\times$ area.

The distractors. Every wrong option is a real product of the same three symbols, so each one has to be checked rather than dismissed.

- **$\rho \, v \, \text{squared} \, d$** gives $M \, T^{-2}$. The squared speed has put a second time into the denominator, so this is a force per unit length rather than a mass per second. It is the nearest miss of the four: only one exponent differs from the answer.

- **$\rho \, v / d$** gives $M \, L^{-3} \, T^{-1}$. Dividing by a length where the answer needs a length squared leaves three powers of the length in the denominator.

- **$\rho \, v \, d$** gives $M \, L^{-1} \, T^{-1}$. It is one power of the length short of cancelling — a very easy slip if you count the density's L^{-3} and the speed's L as cancelling and then forget that the length d has to supply two more.

- **$\rho \, v \, \text{squared} \, d \, \text{squared}$** gives $M \, L \, T^{-2}$, which is a force. It overshoots the length by one and the time by one at the same time, so two errors have partly cancelled and the result looks superficially reasonable.

The trap. Treating dimensional analysis as pattern-spotting. The five options differ only in their exponents, so the eye has almost nothing to work with and the confident answer is usually the one that was seen most recently. The method is mechanical and cannot be faked: one equation per base dimension, solved in the order mass, time, length, because that is the order in which the equations become solvable.

Relevant topics: base dimensions; writing a quantity as a product of powers; matching exponents dimension by dimension; mass flow rate; checking a derived combination.

The trap. Reading the combination off the list by eye. Three base dimensions give three equations, and two of them name an exponent outright, so the answer is forced rather than chosen.

Q2 · A hinged rod held by a cable: the tension from moments — **Answer C**

What is being tested. Whether you can take moments about a hinge to dispose of the unknown reaction, and whether you resolve the cable correctly -- only its vertical part turns the rod about A.

Step 1 — the cable's direction. The rod is 4.0 m out, the cable rises 3.0 m to the wall, so the cable, the rod and the wall make a 3-4-5 triangle. The tension therefore has a vertical component $T_y = T \times 3/5$ and a horizontal component $T \times 4/5$.

Step 2 — choose the pivot. About the hinge A the two hinge reactions have zero moment arm, so they vanish from the equation. The only unknown left is T.

Step 3 — the clockwise moments. The 150 N load sits at the midpoint, 2.0 m from A : moment $150 \times 2.0 = 300 \text{ N m}$. The 45 N load hangs at B, 4.0 m from A : moment $45 \times 4.0 = 180 \text{ N m}$. Total clockwise = 480 N m .

Step 4 — the anticlockwise moment. Only the vertical component of the cable turns the rod, and it acts at B (4.0 m from A):

$$T_y \times 4.0 = (3T/5) \times 4.0 = 12T/5$$

Step 5 — solve.

$$\begin{aligned} 12T/5 &= 480 \\ T &= 480 \times 5 / 12 = 200 \text{ N} \end{aligned}$$

So **Answer: C**.

Sanity check. The cable pulls at an angle, so its full tension must exceed the 150 N and 45 N loads that it balances between them; 200 N is just above that, which is the right direction. The 5.0 m cable length never appears, because only the 3.0 m rise does any vertical work.

The distractors.

- **150 N** is the mid-rod load on its own -- a candidate who balances only that and forgets the end load and the cable angle lands here.
- **250 N** uses the full 5.0 m cable as the tension's moment arm, treating the cable as if it pulled straight down.
- **120 N** divides the total moment by the 4.0 m rod but omits the 3/5 vertical factor, so it under-values the cable.
- **160 N** divides by the horizontal arm but forgets to multiply the vertical component by the 4.0 m distance from the hinge.

Relevant topics: moments about a pivot; resolving a tension; the 3-4-5 triangle; why the hypotenuse is a distraction.

The trap. Using the 5.0 m cable length as the moment arm of the tension. Only the 3.0 m vertical rise (the 3/5 component) does any turning about the hinge; the hypotenuse never enters the equation.

Q3 · Two potential dividers: the voltmeter reading between their midpoints — **Answer E**

What is being tested. Whether you treat each branch as an independent potential divider and then take the difference of two internal potentials -- not a drop across a single resistor.

Step 1 — the left branch current. Its total resistance is $2.0 + 6.0 = 8.0$ ohm, so

$$I_{\text{left}} = 12 / 8.0 = 1.5 \text{ A}$$

Step 2 — the potential of X. X is the midpoint; below it sits the 6.0 ohm resistor, across which the drop is $1.5 \times 6.0 = 9.0$ V. So X is 9.0 V above the bottom rail, equivalently $12 - 9.0 = 3.0$ V below the top -- but the relevant number is $V_X = 9.0$ V above the rail.

Step 3 — the right branch current. Its total resistance is $6.0 + 3.0 = 9.0$ ohm :

$$I_{\text{right}} = 12 / 9.0 = 4/3 \text{ A}$$

Step 4 — the potential of Y. Below Y sits the 3.0 ohm resistor, drop $(4/3) \times 3.0 = 4.0$ V. So $V_Y = 4.0$ V above the bottom rail.

Step 5 — the meter reading. An ideal voltmeter draws no current, so it simply reads the potential difference between X and Y:

$$V = V_X - V_Y = 9.0 - 4.0 = 5.0 \text{ V}$$

So **Answer: E.**

The distractors.

- **3.0 V** is the drop across the 2.0 ohm resistor on the left branch -- the distance from X up to the top rail, not the meter reading.
- **8.0 V** is the drop across the 6.0 ohm resistor on the right branch.
- **9.0 V** is the potential of X above the rail; a candidate who forgets to subtract Y reads the supply-split instead of the difference.
- **1.5 V** is merely the left branch current, written as though it were a voltage.

Relevant topics: potential dividers; branch currents; an ideal voltmeter reads a potential difference.

The trap. Reading the drop across a single resistor (the 2.0 ohm or the 6.0 ohm) instead of the potential difference X minus Y. The meter bridges two internal points, so both midpoints must be found first and then subtracted.

Q4 · Order of magnitude: the number of seconds in one year — **Answer B**

What is being tested. A clean order-of-magnitude estimate: round each factor sensibly and read the power of ten, without a calculator.

Step 1 — express a year in seconds.

$$\begin{aligned}
 1 \text{ year} &= 365 \text{ days} \times 24 \text{ hours} \times 3600 \text{ seconds} \\
 &= 31536000 \text{ s}
 \end{aligned}$$

Step 2 — round to one significant figure. The product is about 3.2×10^7 seconds. The nearest power of ten to 3.2×10^7 is 10^7 (it would need to pass $3.2 \times 10^7 \rightarrow 10^{7.5}$ to round the other way). So **Answer: B**.

The distractors.

- 10^5 is the order of the number of seconds in a day (86400), not a year.
- 10^6 is the number of seconds in an hour, a thousand times too small.
- 10^8 is about a hundred times too large, as if a year were closer to a century.
- 10^9 is a thousand times too large, as if the year were a millennium.

Relevant topics: order-of-magnitude estimation; powers of ten; the pi-times-ten-to-the-seven seconds-per-year rule.

The trap. Confusing the scales of a day (10^5 s), an hour (10^6 s) and a year (10^7 s). The day and the hour are the two that tempt a candidate who multiplies wrongly.

Q5 · A velocity-time graph: the total distance from the area — **Answer D**

What is being tested. The single most-used fact about a velocity-time graph: the area beneath it is the distance, read off the drawn axes.

Step 1 — identify the three regions. From the graph: velocity rises linearly from 0 to 20 m/s over the first 5 s; stays at 20 m/s until 15 s; falls linearly to 0 by 19 s. So the area is a triangle, then a rectangle, then a triangle.

Step 2 — the first triangle (0 to 5 s). Its area is $A1 = \frac{1}{2} \times 5 \times 20 = 50 \text{ m}$.

Step 3 — the rectangle (5 to 15 s). The cruise lasts $15 - 5 = 10 \text{ s}$ at 20 m/s:

$$A2 = 10 \times 20 = 200 \text{ m}$$

Step 4 — the second triangle (15 to 19 s). The braking phase lasts $19 - 15 = 4 \text{ s}$:

$$A3 = \frac{1}{2} \times 4 \times 20 = 40 \text{ m}$$

Step 5 — add them.

$$\text{distance} = 50 + 200 + 40 = 290 \text{ m}$$

So **Answer: D**.

The distractors.

- **250 m** double-counts the braking triangle or tacks it onto the rectangle twice.

- **300 m** treats the whole 15 s as a 20 m/s rectangle (15 x 20) and ignores that the first and last phases are triangles.
- **200 m** is the cruise rectangle only.
- **340 m** invents an extra triangle that is not on the graph.

Relevant topics: v-t graphs; area = distance; splitting into triangles and rectangles.

The trap. Taking the cruise rectangle (200 m) as the whole answer and ignoring the two triangular end regions. Each triangle is real distance, and only their sum is the journey.

Q6 · A ball down a smooth ramp, across a rough floor, then sticking to a second ball: how far the combined mass slides — **Answer D**

What is being tested. Whether you notice that this is three problems in a row, and that each one obeys a different rule. Energy is conserved on the smooth ramp. Energy is *not* conserved along the rough floor — it leaks away as heat. And at the collision energy is not conserved at all, while momentum is.

Step 1 — what the ramp delivers. The ramp is smooth, so no energy is lost on it and all the potential energy the ball had at the top becomes kinetic energy by the bottom:

$$E = m g h = 0.50 \times 10 \times 0.40 = 2.0 \text{ J}$$

That 2.0 J is the whole energy budget for the rest of the question. Everything that follows is a subtraction from it.

Step 2 — the friction force on the first ball. The rough surface begins at the foot of the ramp. The friction force on a horizontal surface is

$$F = \mu m g = 0.20 \times 0.50 \times 10 = 1.0 \text{ N}$$

Note that only the 0.50 kg ball is moving at this stage, so only 0.50 kg belongs in this force.

Step 3 — the work it does over the 1.0 m.

$$W = F d = 1.0 \times 1.0 = 1.0 \text{ J}$$

Step 4 — the energy the ball still has at the collision.

$$E = 2.0 - 1.0 = 1.0 \text{ J}$$

Half the energy is already gone before anything is hit. This is the step that is most often skipped: the 1.0 m of rough floor is easy to read past.

Step 5 — the speed at the collision. From the kinetic energy,

$$\begin{aligned} \text{one half } m v^2 &= 1.0 \\ v^2 &= 2 \times 1.0 / 0.50 = 4.0 \\ v &= 2.0 \text{ m/s} \end{aligned}$$

Step 6 — the momentum before the collision.

$$p = m v = 0.50 \times 2.0 = 1.0 \text{ kg m/s}$$

Step 7 — the collision. The two balls stick, so this is perfectly inelastic. Momentum is conserved; kinetic energy is not. The combined mass is 0.80 kg:

$$\begin{aligned} 0.80 \times V &= 1.0 \\ V &= 1.25 \text{ m/s} \end{aligned}$$

It is worth pausing on how much energy has just vanished. Before the collision the kinetic energy was 1.0 J; afterwards it is one half $\times 0.80 \times 1.25$ squared, which is 5/8 J. The collision destroyed 3/8 J — more than a third of what was left. That energy went into heating and deforming the balls, and it is not available to push them along the floor.

Step 8 — the friction on the combined mass. The mass has grown, so the friction force has grown with it:

$$F = \mu m g = 0.20 \times 0.80 \times 10 = 1.6 \text{ N}$$

Step 9 — the distance. All the remaining kinetic energy is turned into work against that force:

$$d = E / F = (5/8) / 1.6 = 25/64 = 0.39 \text{ m}$$

So **Answer: D.**

Step 10 — check the answer against the two effects. Both changes after the collision push the same way. The collision threw away 3/8 J of the 1.0 J that was left, and the friction force rose from 1.0 N to 1.6 N. So the slide must be shorter than the 1.0 m that came before, and it is. As a second check, ask what would happen with no collision loss at all: 1.0 J against 1.6 N would give 0.63 m, so the collision alone costs about 0.24 m, and the heavier friction costs a little more. 0.39 m is consistent with both.

The distractors.

- **0.63 m** carries the 1.0 J through the collision unchanged. It is the most dangerous wrong answer, because every number in it is right except one assumption: that a collision which joins two bodies together leaves their kinetic energy alone. Joining two bodies always destroys kinetic energy, and the amount destroyed here is a third of the total.
- **0.78 m** forgets the 1.0 m of rough floor before the collision, so the ball arrives with the full 2.0 J. It gets 1.25 J after the collision and divides by 1.6 N. The rough stretch is drawn on the figure precisely because it is easy to miss.
- **0.16 m** uses the combined 0.80 kg when working out the friction on the first stretch. At that moment only the 0.50 kg ball is moving, and the second ball is at rest and has no friction to overcome.
- **0.31 m** comes from writing the kinetic energy after the collision as one half $m V$ rather than one half $m V$ squared. That gives 0.5 J instead of 0.625 J, and the distance falls to 0.31 m. It is a slip of one symbol and it costs 20 per cent of the answer.

The trap. Applying one conservation law across a whole question that needs three. The give-away is the word "stick": wherever two bodies join, momentum carries you across the collision and kinetic energy does not. And wherever a surface is rough, energy is leaving the account entirely, so it must be subtracted rather than conserved.

Relevant topics: conservation of energy on a smooth surface; work done against friction; kinetic energy and speed; conservation of momentum in a perfectly inelastic collision; multi-stage problems.

The trap. Using one rule for all three stages. Momentum carries the motion through the collision, not kinetic energy, and the 1.0 m of rough floor before the collision has already taken half the energy away.

Q7 · A vertical circle: the minimum speed at the bottom for the string to stay taut — **Answer A**

What is being tested. The classic vertical-circle condition: the string stays taut only if, at the top, gravity alone can supply the centripetal force. Energy then connects that top speed to the launch speed at the bottom.

Step 1 — the taut-string condition at the top. At the highest point the tension is smallest. The string is just taut when $T = 0$, so weight provides all the centripetal force:

$$m v_{\text{top}}^2 / r = m g \quad \Rightarrow \quad v_{\text{top}}^2 = g r$$

Step 2 — energy from bottom to top. The top is $2r$ higher than the bottom, so with no work lost:

$$\frac{1}{2} m u^2 = \frac{1}{2} m v_{\text{top}}^2 + m g (2r)$$

Step 3 — substitute and solve for u. Divide by m and put in $v_{\text{top}}^2 = g r$:

$$\frac{1}{2} u^2 = \frac{1}{2} g r + 2 g r = \left(\frac{1}{2} + 2\right) g r = \left(\frac{5}{2}\right) g r$$

$$u^2 = 5 g r$$

$$u = \sqrt{5 g r}$$

So **Answer: A**.

Step 4 — check the size. The answer must exceed both $\sqrt{g r}$ (the top speed) and $\sqrt{4 g r}$ (the speed from a $2r$ drop alone), because the particle needs enough speed to climb the two radii AND keep the string taut. $\sqrt{5 g r}$ does that.

The distractors.

- $\sqrt{2 g r}$ uses only the energy gain $2 g r$ as the speed, forgetting that the top still needs $g r$ of centripetal speed.
- $\sqrt{3 g r}$ counts three radii of rise instead of two.
- $\sqrt{g r}$ is the top speed with the particle momentarily at rest there -- which is exactly when the string goes slack.
- $\sqrt{4 g r}$ is the speed gained in falling $2r$, ignoring the centripetal requirement at the top.

Relevant topics: vertical circular motion; the taut-string condition; conservation of mechanical energy.

The trap. Forgetting that at the top the particle still needs $v_{\text{top}}^2 = g r$ for the string to stay taut, and giving $\sqrt{2gr}$ or $\sqrt{4gr}$ from energy alone. The $5gr$ combines a $2r$ climb with the top's own centripetal need.

Q8 · A wave snapshot: the transverse velocity from the local slope — **Answer C**

What is being tested. The link between a snapshot (which shows slope) and motion (which is velocity): for a right-travelling wave the transverse velocity is minus the wave speed times the local slope.

Step 1 — the velocity-slope relation. A wave moving right has the form $y = f(x - vt)$.

Differentiating with respect to time at fixed x :

$$dy/dt = -v (dy/dx)$$

so the transverse velocity is the wave speed times the slope, with a minus sign.

Step 2 — read the slope at Q. The stem gives the slope at Q as -0.40 . In the snapshot Q lies on the descending flank between the crest and the trough, so the string falls as x increases -- a negative slope, as stated.

Step 3 — compute.

$$v_y = -1.5 \times (-0.40) = +0.60 \text{ m/s}$$

A positive transverse velocity is upward. So **Answer: C**.

Step 4 — check the direction. On a right-travelling wave a point on the descending side is being lifted toward the crest that is approaching from the left, so it moves up -- consistent with the positive sign.

The distractors.

- **0.60 m/s downward** keeps the magnitude but drops the minus sign of the relation, sending the point the wrong way.
- **0.40 m/s upward** multiplies the speed by the slope with no minus, then loses the factor of 1.5.
- **0.90 m/s upward** adds the speed and the slope as if they were a sum.
- **0 m/s** is the velocity at a crest or a trough, where the slope is zero, not at Q.

Relevant topics: transverse waves; $v_y = -v (dy/dx)$; reading slope from a snapshot.

The trap. Dropping the minus sign in $v_y = -v(dy/dx)$, so a downward-sloping point (Q, on the descending flank) is sent downward instead of upward. The sign is the entire question.

Q9 · A converging lens: an object inside the focal length gives a virtual image — **Answer E**

What is being tested. The lens equation when the object is inside the focal length -- the case that yields a virtual, magnified image -- and the sign convention that makes v come out negative.

Step 1 — the lens equation. With the usual convention (real object and image distances positive, converging lens f positive):

$$1/u + 1/v = 1/f$$

Step 2 — substitute. Here $u = 12 \text{ cm}$ and $f = 20 \text{ cm}$. Note $u < f$, so the object is inside the focal length.

$$1/v = 1/f - 1/u = 1/20 - 1/12$$

Step 3 — the arithmetic.

$$1/20 - 1/12 = 3/60 - 5/60 = -2/60 = -1/30$$

$$\text{so } v = -30 \text{ cm}$$

The negative sign means the image is virtual and on the same side of the lens as the object.

Step 4 — the magnification.

$$m = -v/u = -(-30)/12 = 30/12 = 2.5$$

So Answer: E.

Step 5 — check. A virtual image is upright (m positive) and, for an object inside the focus, enlarged ($m > 1$). 2.5 satisfies both, which is the right physical sense.

The distractors.

- **0.40** is u/v in magnitude, the reciprocal of the magnification -- the answer turned upside down.
- **1.7** is $20/12$, the ratio of focal length to object distance, which is not the magnification.
- **2.0** comes from a mis-rearranged equation such as $f/(f - u)$.
- **0.60** is $12/20$, the ratio of object distance to focal length.

Relevant topics: the lens equation; the virtual image for $u < f$; magnification $m = -v/u$.

The trap. Swapping u and v to get 0.40, or using f/u and u/f , when the magnification is $-v/u$ with a virtual (negative) v . The negative v is the signal that the image is enlarged, not diminished.

Q10 · Two opposing cells: the current in the single loop — **Answer B**

What is being tested. Two cells facing opposite ways subtract, and internal resistances add to the external resistor. The loop is then one net emf over one total resistance.

Step 1 — the net emf. The 12 V cell has its positive terminal up and the 6 V cell has its positive terminal down, so around the loop they drive in opposite senses. The 12 V cell is larger, so:

$$E_{\text{net}} = 12 - 6 = 6 \text{ V}$$

Step 2 — the total resistance. The resistor and both internal resistances are in series:

$$R_{\text{total}} = 3.0 + 1.0 + 2.0 = 6.0 \text{ ohm}$$

Step 3 — the current.

$$I = E_{\text{net}} / R_{\text{total}} = 6 / 6.0 = 1.0 \text{ A}$$

Step 4 — the voltage shares. The 6.0 ohm total means each ohm carries the 1.0 A, so the two internal resistances ($1.0 + 2.0 = 3.0$ ohm) drop $3.0 \times 1.0 = 3.0 \text{ V}$, leaving $6.0 - 3.0 = 3.0 \text{ V}$ across the external 3.0 ohm resistor -- consistent with $3.0 \times 1.0 = 3.0 \text{ V}$.

$$V_{\text{internal}} = (1.0 + 2.0) \times 1.0 = 3.0 \text{ V} ; V_{\text{load}} = 3.0 \times 1.0 = 3.0 \text{ V}$$

So Answer: B.

Step 5 — check the sense. The net emf is positive in the direction of the 12 V cell, so the current circulates that way, and its modest size (1.0 A) reflects that the two cells are largely cancelling each other.

The distractors.

- **2.0 A** drives both emfs the same way: $(12 + 6)/9 = 2.0$ -- adding when they oppose.
- **3.0 A** adds the emfs and uses all three resistances but with the wrong combination $18/6$.
- **0.50 A** divides the net 6 V by only the external 3.0 ohm, forgetting the internal resistances.
- **6.0 A** mistakes the net emf of 6 V for the current in amperes.

Relevant topics: cells in series opposing; internal resistance; Kirchhoff's loop law.

The trap. Adding the two emfs because both are batteries, instead of subtracting them, or leaving out the internal resistances. The net drive is $12 - 6 = 6$ V over 6.0 ohm total.

Q11 · Radioactive decay: the fraction remaining after a number of half-lives — **Answer B**

What is being tested. The geometric nature of radioactive decay: after each half-life the remaining amount is halved, so after n half-lives it is $(1/2)^n$.

Step 1 — how many half-lives?

$$n = 24 \text{ days} / 8.0 \text{ days} = 3$$

Step 2 — apply the halving three times.

$$\text{fraction} = (1/2)^3 = 1/8$$

Step 3 — follow the decay one step at a time. Rather than jump to the cube, watch each halving: after the first half-life the fraction is $1/2$, after the second it is $(1/2) \times (1/2) = 1/4$, after the third it is $(1/4) \times (1/2) = 1/8$.

$$1/2 \rightarrow 1/4 \rightarrow 1/8 \text{ over three halvings}$$

Step 4 — confirm the time divides exactly. 24 days is three exact 8.0-day intervals with no remainder, so there is no partial fourth half-life to account for; the fraction is the clean value $(1/2)^3$.

So **Answer: B.**

Step 5 — check the bracket. Three halvings must give something below $1/2$ (one halving) but above $1/16$ (four halvings). $1/8$ sits exactly there.

The distractors.

- **$1/4$** is the result of only two half-lives, as if 24/8 had been read as two.
- **$1/16$** is four half-lives -- one too many.
- **$3/8$** subtracts the three half-lives from 1, treating decay as a linear loss.
- **$1/2$** is the fraction after a single half-life, the start of the process rather than its end.

Relevant topics: half-life; geometric decay; the $(1/2)^n$ rule.

The trap. Treating the three half-lives as a subtraction from 1 (giving $3/8$) instead of three multiplications by $1/2$. Decay is geometric, so each half-life multiplies the remainder by $1/2$.

Q12 · The momentum of a photon of given wavelength — **Answer D**

What is being tested. The photon momentum formula $p = h/\lambda$, and keeping it distinct from the photon energy $E = hc/\lambda$.

Step 1 — the formula. A photon of wavelength λ has momentum

$$p = h / \lambda$$

Step 2 — the wavelength in metres.

$$600 \text{ nm} = 600 \times 10^{-9} \text{ m} = 6.0 \times 10^{-7} \text{ m}$$

Step 3 — substitute.

$$p = 6.6 \times 10^{-34} / (6.0 \times 10^{-7}) = 1.1 \times 10^{-27} \text{ kg m s}^{-1}$$

So **Answer: D**.

Step 4 — dimensional check. $(\text{J s})/\text{m} = (\text{kg m}^2 \text{ s}^{-2} \text{ s})/\text{m} = \text{kg m/s}$, which is a momentum. That confirms the formula, not the energy.

The distractors.

- 3.3×10^{-19} is the photon energy hc/λ , the wrong quantity.
- 5.5×10^{-28} loses a factor of ten in the nanometre-to-metre conversion.
- 1.1×10^{-19} uses the energy hf without dividing by c , massively too large.
- 6.6×10^{-34} is just Planck's constant with the wavelength ignored.

Relevant topics: photon momentum; $p = h/\lambda$; the difference between photon energy and momentum.

The trap. Computing the photon energy hc/λ and offering it as the momentum, or forgetting to convert 600 nm to 6.0×10^{-7} m. Momentum is h/λ , not hc/λ .

Q13 · Converting a speed from m/s to km/h — **Answer A**

What is being tested. A two-step unit conversion: get the speed in m/s, then change to km/h with the factor 3.6.

Step 1 — the speed in m/s.

$$v = 100 \text{ m} / 10 \text{ s} = 10 \text{ m/s}$$

Step 2 — convert to km/h. One metre per second is $(1/1000 \text{ km})/(1/3600 \text{ h}) = 3.6 \text{ km/h}$. So:

$$v = 10 \times 3.6 = 36 \text{ km/h}$$

So **Answer: A.**

The distractors.

- **10 km/h** is the speed in m/s left unconverted.
- **3.6 km/h** divides by 3.6 instead of multiplying.
- **360 km/h** multiplies the 100 and 10 together as if they made the speed.
- **100 km/h** offers the distance as the speed.

Relevant topics: average speed; the m/s to km/h conversion factor 3.6.

The trap. Leaving the answer as 10 m/s, or dividing by 3.6 instead of multiplying. From m/s to km/h you multiply by 3.6 (1000 m per km, 3600 s per hour).

Q14 · A horizontal projectile: the range from the time of fall — **Answer C**

What is being tested. The independence of horizontal and vertical motion for a projectile launched horizontally: the fall time comes from the height, the range from the horizontal speed.

Step 1 — the time of fall. Vertically the ball starts with no speed and drops 20 m under gravity:

$$h = \frac{1}{2} g t^2 \quad \Rightarrow \quad t = \sqrt{2h/g} = \sqrt{2 \times 20 / 10} = \sqrt{4} = 2.0 \text{ s}$$

Step 2 — the range. Horizontally the speed stays 12 m/s, so in 2.0 s the ball travels

$$\text{range} = 12 \times 2.0 = 24 \text{ m}$$

Step 3 — why the fall time also sets the horizontal distance. Horizontal and vertical motion are independent: the 12 m/s launch does nothing to how fast the ball falls, so the 2.0 s it takes to drop is the same 2.0 s available for horizontal travel. That is what licenses the relation range = speed x time.

$$\text{horizontal distance} = v_x \times t_{\text{fall}} = 12 \times 2.0$$

Step 4 — check the units and the scale. $\sqrt{2h/g}$ has units $\sqrt{\text{m} / (\text{m/s}^2)} = \sqrt{\text{s}^2} = \text{s}$, so the time is dimensionally sound; and a 20 m drop yielding a 2 s fall, stretched by 12 m/s, lands at 24 m -- the right order.

So **Answer: C.**

Step 5 — check the answer. The range should exceed the 20 m height (it does) and be of order the speed times a couple of seconds. 24 m fits.

The distractors.

- **12 m** uses the 12 m/s launch speed as if it were the time of fall.
- **48 m** doubles the correct range, perhaps using $2t$ or adding the height.
- **34 m** is a muddled blend that sits between 24 and 48.
- **17 m** takes $\sqrt{2h/g}$ as the range and never multiplies by the horizontal speed.

Relevant topics: horizontal projectile; independent components; range = $v_x t$.

The trap. Using the 12 m/s launch speed as the time of fall, or doubling the result. The time of fall is $\sqrt{2h/g} = 2.0$ s, and the range is horizontal speed (12 m/s) times that time.

Q15 · Two perpendicular forces: the equilibrating force — **Answer E**

What is being tested. Combining two perpendicular forces by Pythagoras, then realising the equilibrium force is equal and opposite to their resultant.

Step 1 — the resultant of the two given forces. They are at right angles, so the resultant R has

$$R^2 = 3^2 + 4^2 = 9 + 16 = 25$$

$$R = 5 \text{ N}$$

Step 2 — the equilibrium force. For the body to be in equilibrium the vector sum of all forces must be zero, so the third force must cancel R: same magnitude, opposite direction.

required force = 5 N, opposite to the resultant

Step 3 — resolve the two forces onto axes. Put the 3 N force along x and the 4 N force along y. Their resultant then has components $R_x = 3$ N and $R_y = 4$ N -- a vector sum, not an arithmetic one.

Step 4 — recover the magnitude from the components. Because the components are perpendicular, Pythagoras gives the magnitude:

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{3^2 + 4^2} = 5 \text{ N}$$

So **Answer: E.**

Step 5 — check the size. A vector sum of 3 N and 4 N must exceed each but fall short of their sum 7 N. 5 N does exactly that.

The distractors.

- **7 N** adds the forces as if parallel ($3 + 4$), ignoring the right angle.
- **1 N** subtracts them ($4 - 3$), as if they opposed along one line.
- **12 N** multiplies them, which is not even a force dimensionally.
- **3.5 N** averages them, another way to ignore the geometry.

Relevant topics: resultant of perpendicular forces; Pythagoras; equilibrium as zero vector sum.

The trap. Adding the two forces as parallel (7 N) or subtracting them (1 N) instead of combining them as vectors at right angles. The resultant is $\sqrt{3^2 + 4^2} = 5$ N, and the equilibrant is 5 N opposite.

Q16 · A rotating sphere: how latitude changes the centripetal acceleration — **Answer E**

What is being tested. That centripetal acceleration depends on the distance from the rotation axis, which for a point at latitude λ is $R \cos(\lambda)$, not R.

Step 1 — the radius of the circular path. The sphere spins about its vertical axis. A point at latitude λ is on a circle whose radius is its distance from that axis:

$$r_{\text{circle}} = R \cos \lambda$$

(At $\lambda = 0$, the equator, this is R ; at the poles it is 0.)

Step 2 — the centripetal acceleration.

$$a = \omega^2 r_{\text{circle}} = \omega^2 R \cos \lambda$$

On the equator the acceleration is $\omega^2 R$. The ratio is therefore

$$\text{ratio} = (\omega^2 R \cos \lambda) / (\omega^2 R) = \cos \lambda$$

Step 3 — derive the latitude radius from the geometry. At latitude λ the point sits at height $R \sin \lambda$ above the equatorial plane, so by Pythagoras its horizontal distance from the axis is $\sqrt{R^2 - (R \sin \lambda)^2} = R \cos \lambda$. That is why the equator ($\lambda = 0$) gives R and the poles give 0 -- the geometry, not a guess.

$$r_{\text{circle}} = \sqrt{R^2 - R^2 \sin^2 \lambda} = R \cos \lambda$$

Step 4 — apply the centripetal formula to that radius. Since $a = \omega^2 r_{\text{circle}}$, substituting the latitude radius gives $a = \omega^2 R \cos \lambda$; dividing by the equatorial value $\omega^2 R$ leaves $\cos \lambda$, with no extra factor.

So Answer: E.

Step 5 — check the endpoints. At the equator ($\lambda = 0$) the factor is $\cos 0 = 1$; at the poles ($\lambda = 90$) it is $\cos 90 = 0$. Cosine is the only option that does both, and at 45 degrees it gives $\sqrt{2}/2$ (about 0.71).

The distractors.

- **$\sin \lambda$** uses the wrong trigonometric factor for the distance from the axis.
- **$\cos^2 \lambda$** squares the correct factor, double-counting the latitude dependence.
- **$\sin^2 \lambda$** is the wrong factor squared.
- **1** claims the acceleration is the same everywhere, ignoring latitude entirely.

Relevant topics: circular motion on a rotating sphere; the radius of the latitude circle; centripetal acceleration and latitude.

The trap. Using R instead of $R \cos(\lambda)$ for the radius of the latitude circle, or $\sin(\lambda)$, or squaring. The distance from the spin axis is $R \cos(\lambda)$, so the acceleration scales as $\cos(\lambda)$: 1 at the equator, 0 at the poles.

Q17 · A square lamina with a corner removed: the shift of the centre of mass — **Answer B**

What is being tested. The negative-mass method for a centre of mass: the cut-out is subtracted from the whole, and the remaining lamina is the difference, weighted by area (which stands in for mass for a

uniform sheet).

Step 1 — set up coordinates. Let the original square run from $(0,0)$ to $(2a, 2a)$, so its centre is at (a, a) and its area is $4a^2$. The removed upper-left corner runs from $(0, a)$ to $(a, 2a)$, area a^2 , centre at $(a/2, 3a/2)$.

Step 2 — the remaining lamina. Its area is $4a^2 - a^2 = 3a^2$. By the negative-mass idea:

$$x_{\text{new}} = (4a^2 \times a - a^2 \times a/2) / (3a^2)$$

Step 3 — the horizontal position. Cancel a^2 :

$$x_{\text{new}} = (4a - a/2) / 3 = (7a/2) / 3 = 7a/6$$

Step 4 — the displacement. The original centre was at $x = a$, so the shift is

$$\text{shift} = 7a/6 - a = a/6$$

to the right, toward the remaining material. So **Answer: B**.

Step 5 — check. The cut is a quarter of the area at the upper left, so the centre of mass should move a little to the right and down -- $a/6$ in x is small and in the right direction. (The vertical shift is $a/6$ down as well, by symmetry of the cut.)

The distractors.

- **a/3** treats the removed corner as a third of the mass rather than a quarter.
- **a/4** confuses the side of the cut-out with the displacement.
- **a/2** is the half-side of the whole lamina, not a displacement at all.
- **a/12** double-reduces the moment and lands a quarter of the $a/3$ error.

Relevant topics: centre of mass; the negative-mass (cut-out) method; uniform lamina as area.

The trap. Treating the removed corner as a third of the mass, or confusing the cut-out's side with the displacement. The corner is a quarter of the area, and the new centre of mass is (original moment minus corner moment) over the remaining three-quarter mass, giving a shift of $a/6$.

Q18 · Two coherent sources: the fringe spacing — **Answer D**

What is being tested. The two-source fringe spacing $\text{beta} = \lambda D / d$, and the discipline of converting every length to the same unit first.

Step 1 — the formula.

$$\text{beta} = \lambda D / d$$

Step 2 — consistent units (metres).

$$\begin{aligned}\lambda &= 500 \text{ nm} = 5.0 \times 10^{-7} \text{ m} \\ d &= 1.0 \text{ mm} = 1.0 \times 10^{-3} \text{ m} \\ D &= 1.5 \text{ m}\end{aligned}$$

Step 3 — substitute.

$$\begin{aligned}\beta &= (5.0 \times 10^{-7} \times 1.5) / (3.0 \times 10^{-4}) \\ &= 7.5 \times 10^{-4} \text{ m} \\ &= 0.75 \text{ mm}\end{aligned}$$

So **Answer: D.**

Step 4 — check. For visible light and a millimetre-scale slit spacing the fringes are sub-millimetre, so 0.75 mm is the right order; a value of several millimetres would mean the formula or units were wrong.

The distractors.

- **0.25 mm** inverts the formula to $\lambda d / D$.
- **1.5 mm** offers the screen distance D itself.
- **2.5 mm** is a unit-muddled value from mixing mm and nm.
- **0.50 mm** swaps λ and d, putting the 0.30 mm separation in the numerator with no D.

Relevant topics: two-source interference; $\beta = \lambda D / d$; unit consistency.

The trap. Inverting the fringe-spacing formula ($\lambda d / D$) or forgetting to convert nm and mm to metres. $\beta = \lambda D / d$, and with $\lambda = 5.0 \times 10^{-7} \text{ m}$, $d = 1.0 \times 10^{-3} \text{ m}$, $D = 1.5 \text{ m}$ it is 0.75 mm.

Q19 · The critical angle: its sine for glass to air — **Answer A**

What is being tested. The critical-angle condition and, specifically, that its sine is the reciprocal of the refractive index (n_2/n_1), not the index itself.

Step 1 — the condition. At the critical angle the refracted ray grazes the boundary, so the refracted angle is 90 degrees and Snell's law reads

$$n_1 \sin \theta_c = n_2 \sin 90 = n_2$$

Step 2 — glass to air. Here $n_1 = 1.50$ (glass) and $n_2 = 1.00$ (air):

$$\sin \theta_c = n_2 / n_1 = 1.00 / 1.50 = 2/3$$

Step 3 — the value. $2/3$ is about 0.67. So **Answer: A.**

Step 4 — check. The sine must be below 1 (a critical angle exists only because glass is denser than air) and above 0.5 (the critical angle is below 45 degrees since $1/1.5 > 1/\sqrt{2}$). 0.67 sits correctly between.

The distractors.

- **1.50** is the refractive index read directly as the sine, instead of its reciprocal $2/3$.
- **0.50** would be the sine if the glass index were 2.00 rather than 1.50.
- **0.33** comes from a difference error, $(n_1 - n_2)/n_1 = 1/3$, instead of the ratio n_2/n_1 .
- **1.00** says $\sin(\theta_c) = 1$, which would mean there is no critical angle -- but glass to air does have one.

Relevant topics: total internal reflection; the critical angle; $\sin(\theta_c) = n_2/n_1$.

The trap. Giving the refractive index 1.50 directly, or 0.50 (as if the index were 2.0), or 1/3 from a difference error, or 1.00. The sine is $n_2/n_1 = 2/3$, the reciprocal of the index.

Q20 · A diode and a load resistor: the operating current from the load line — **Answer C**

What is being tested. The load-line construction: a resistor and battery impose a straight line on the diode's I-V plane, and the operating current is where that line meets the device curve -- not a point read off either alone.

Step 1 — the load line. The supply is 1.0 V and the resistor is 100 ohm. In mA, a voltage V across the diode leaves $1.0 - V$ across the resistor, carrying current $(1.0 - V)/100$ amps = $10(1.0 - V)$ mA. So

$$I_{\text{mA}} = 10 (1.0 - V) \Rightarrow V = 1.0 - 0.1 I_{\text{mA}}$$

This line runs from $(V=1.0, I=0)$ to $(V=0, I=10 \text{ mA})$.

Step 2 — the diode segment. Between 0.55 V and 0.82 V the characteristic is straight and passes through 22 mA at 0.8 V. Its slope is

$$I_{\text{mA}} = 22 (V - 0.55) / (0.82 - 0.55) = 22 (V - 0.55) / 0.27$$

Step 3 — intersect. Substitute $V = 1.0 - 0.1 I_{\text{mA}}$ into the diode line:

$$\begin{aligned} I_{\text{mA}} &= 22 (1.0 - 0.1 I_{\text{mA}} - 0.55) / 0.27 \\ &= 22 (0.45 - 0.1 I_{\text{mA}}) / 0.27 \end{aligned}$$

Multiply out: $0.27 I_{\text{mA}} = 9.9 - 2.2 I_{\text{mA}}$, so $(0.27 + 2.2) I_{\text{mA}} = 9.9$, and

$$I_{\text{mA}} = 9.9 / (0.27 + 2.2) \approx 4.0 \text{ mA}$$

So **Answer: C.**

Step 4 — check. At 4.0 mA the resistor drops $0.1 \times 4.0 = 0.4 \text{ V}$, so the diode has about 0.6 V across it -- on the steep part of the curve, consistent with the drawing. The current is well below the 10 mA intercept, as it must be.

The distractors.

- **10 mA** is the short-circuit intercept of the load line, where the diode voltage is forced to zero.
- **22 mA** is the point on the diode curve at 0.8 V, read with no regard for the resistor.
- **2.0 mA** is the current the load line would allow at 0.8 V across the diode ($(1.0-0.8)/100 = 2 \text{ mA}$), not the intersection.
- **6.0 mA** is a value between the two intercepts that is not the true meeting point.

Relevant topics: load-line analysis; the diode I-V characteristic; solving the operating point.

The trap. Reading the 22 mA point off the diode curve, or the 10 mA short-circuit intercept, as the answer. The current is set by the intersection of the load line $V = 1.0 - 0.1 I_{\text{mA}}$ with the diode segment, which is about 4 mA.

Q21 · A parallel pair in series with a third capacitor: the p.d. across the single one, from the charge the whole chain carries — **Answer A**

What is being tested. Whether you read the circuit before you calculate. Two capacitors here add, and the third does not, and getting the two rules the wrong way round is the whole trap. Nothing here needs a calculator: every number divides cleanly.

Step 1 — read the topology. The 2.0 μF and the 4.0 μF are connected in parallel with each other: their left-hand plates meet at one node and their right-hand plates meet at another, so they sit across the same potential difference. That pair is then in series with the 3.0 μF , because every coulomb that leaves the pair has to pass through the 3.0 μF on its way back to the supply.

Step 2 — replace the parallel pair by one capacitor. Parallel capacitors add, because both sit across the same p.d. and their charges simply add:

$$C_{\text{pair}} = 2.0 + 4.0 = 6.0 \mu\text{F}$$

Step 3 — combine that with the 3.0 μF in series. Series capacitors combine as reciprocals, so the network is smaller than either part:

$$\begin{aligned} 1/C &= 1/6.0 + 1/3.0 = 1/6.0 + 2/6.0 = 3/6.0 \\ C &= 2.0 \mu\text{F} \end{aligned}$$

Step 4 — the charge the supply delivers. The whole network is across 12 V, so

$$Q = C V = 2.0 \times 12 = 24 \mu\text{C}$$

Step 5 — put that charge on the 3.0 μF capacitor. This is the step that decides the question. A series chain carries one charge: the pair and the single capacitor are one after the other in the same loop, so the same 24 μC sits on both. Therefore

$$V = Q / C = 24 / 3.0 = 8.0 \text{ V}$$

So **Answer: A.**

Sanity check. The pair carries the same 24 μC , so its p.d. is $24/6.0 = 4.0 \text{ V}$. The two p.d.s must add to the supply: $8.0 + 4.0 = 12 \text{ V}$. They do. And the answer is the larger share, which is what the smaller capacitance should get.

The distractors.

· **4.0 V** is the p.d. across the parallel pair. It is a real quantity in this circuit, it is reached by the same working, and it is the wrong side of the loop.

· **6.0 V** splits the 12 V equally between the pair and the single capacitor. That would be right only if the pair and the single capacitor had the same capacitance, which they do not: 6.0 μF against 3.0 μF .

- **12 V** treats the $3.0\ \mu\text{F}$ as though it were the only component, so that it takes the whole supply. It does not: the pair is in series with it and takes its share.
- **2.0 V** takes the network's total capacitance, $2.0\ \mu\text{F}$, and reports the number as a voltage.

Relevant topics: capacitors in parallel; capacitors in series; the charge carried by a series chain; $Q = CV$ applied to a network and then to one component.

The trap. Applying the wrong rule to the wrong part: adding the pair is right, and adding the $3.0\ \mu\text{F}$ to the pair as well gives $9.0\ \mu\text{F}$ and a nonsense $2.7\ \text{V}$. The chain rule is that series carries one charge, so the $3.0\ \mu\text{F}$ takes all $24\ \mu\text{C}$ and the $6.0\ \mu\text{F}$ pair has to share it.

Q22 · Turning ice at minus ten degrees into water at twenty degrees: three stages, and the time a heater needs —

Answer E

What is being tested. Whether you break a heating problem into stages before you reach for a formula, and whether you can tell the stages that have a temperature change from the one that has none. It is the stage with no temperature change that costs the most.

Step 1 — the stages. The ice has to do three things, in order. It must warm from $-10\ ^\circ\text{C}$ to $0\ ^\circ\text{C}$. It must then melt, at a steady $0\ ^\circ\text{C}$. And the melt water must then warm from $0\ ^\circ\text{C}$ to $20\ ^\circ\text{C}$. Three stages, three separate energy calculations, and only then an addition.

Step 2 — which formula belongs to which stage. A temperature change needs the specific heat capacity, $E = m c \Delta T$. A change of state needs a specific latent heat, $E = m L$, and during it the temperature does not move at all, so there is no ΔT to put in. Getting these two the right way round is most of the question.

Step 3 — warming the ice. The temperature rise is $10\ ^\circ\text{C}$, and the specific heat capacity is the one for *ice*:

$$E = m c \Delta T = 0.50 \times 2100 \times 10 = 10\ 500\ \text{J}$$

Step 4 — melting the ice. The ice is now at $0\ ^\circ\text{C}$ and stays there while it melts, so no temperature appears in this calculation:

$$E = m L = 0.50 \times 3.3 \times 10\ \text{to the fifth} = 165\ 000\ \text{J}$$

Stop and look at that number next to the last one. Melting costs sixteen times as much as warming the ice through ten degrees. The figure is drawn to scale for exactly this reason: the melting bar towers over the other two.

Step 5 — warming the water. The melt water now rises $20\ ^\circ\text{C}$, and the specific heat capacity is the one for *water*, which is twice the value for ice:

$$E = m c \Delta T = 0.50 \times 4200 \times 20 = 42\ 000\ \text{J}$$

Step 6 — the total energy.

$$E = 10\ 500 + 165\ 000 + 42\ 000 = 217\ 500\ \text{J}$$

Step 7 — the time. The heater delivers 500 J every second, so the number of seconds is the total energy divided by 500:

$$t = E / P = 217\,500 / 500 = 435 \text{ s}$$

So **Answer: E.**

Step 8 — check the three terms against each other. The melting term, 165 000 J, is larger than the other two added together, which come to 52 500 J. That is the normal shape of a heating curve: the flat parts, where a change of state is happening, are where the energy goes. A solution in which the temperature-change terms dominate is a solution that has lost a latent heat somewhere.

Step 9 — check the units at each stage. Mass in kilograms, specific heat capacity in joules per kilogram per kelvin, temperature rise in kelvin or degrees Celsius — the same size for a rise — give joules. Specific latent heat in joules per kilogram times kilograms gives joules. Every term is an energy before they are added, so the addition is legitimate and the division by 500 W gives seconds.

Step 10 — check the size of the answer. 435 s is a little over seven minutes. Half a kilogram of ice is a large block, and a 500 W heater is a small one — a hairdryer on low. Seven minutes is the right kind of number, and it is far from both 105 s and 456 s in a way that the everyday feel of the problem can confirm.

The distractors.

- **414 s** starts from ice already at 0 °C and drops the 10 500 J of Step 3. It is a small error with a small effect, and it is the hardest wrong answer to spot because everything else is right.
- **105 s** drops the melting stage altogether, leaving $10\,500 + 42\,000 = 52\,500$ J. This is the classic error: the ice is warmed, the water is warmed, and the change of state in the middle is forgotten even though it costs more than three times the rest. The figure makes the size of the omission plain.
- **456 s** uses the specific heat capacity of water, 4200, for the ice-warming stage, giving 21 000 J instead of 10 500 J. Ice and water are the same substance but not the same material: ice needs only half as much energy per kilogram per degree.
- **351 s** stops at 0 °C and leaves out the final warming to 20 °C, so it uses $10\,500 + 165\,000 = 175\,500$ J. It is the mirror image of 414 s, and it is the more tempting of the two, because stopping once the ice has melted feels like the end of the problem.

The trap. Treating the whole process as a single warming of water. Doing that gives $0.50 \times 4200 \times 30 = 63\,000$ J and a time of 126 s, which is not even one of the options — and it is wrong by a factor of more than three, because it ignores the energy needed to break the ice apart. The defence is to write the three stages down before computing anything, and to ask of each one whether its temperature is changing.

Relevant topics: specific heat capacity; specific latent heat of fusion; the heating curve and its flat portions; energy supplied by a heater; multi-stage energy problems.

The trap. Treating the process as one warming of water and losing the latent heat of fusion, which alone costs more than three times the other two stages put together.

What is being tested. The mass-energy equivalence as a single multiplication, using the convenience that one atomic mass unit converts to 930 MeV.

Step 1 — the principle. Whenever mass is lost, the released energy is

$$E = m c^2$$

Step 2 — the shortcut. Rather than carry c^2 , the problem gives $1 \text{ u } c^2 = 930 \text{ MeV}$. So a defect of 0.20 u gives

$$E = 0.20 \times 930 = 186 \text{ MeV}$$

Step 3 — carry the units through the multiplication. The 930 MeV already bundles c^2 into per u, so the only arithmetic is the mass in u times that rate; keeping the u in view stops the defect (0.20 u) being confused with a whole atom.

$$\text{rate} = 930 \text{ MeV} / \text{u} ; E = 0.20 \text{ u} \times 930 \text{ MeV/u} = 186 \text{ MeV}$$

Step 4 — confirm the figure is consistent. The 930 MeV per u is itself a rounded figure (the exact $1 \text{ u } c^2$ is 931.5 MeV), so 186 MeV is the right magnitude; quoting more digits would overstate the precision of the given constant.

So **Answer: B.**

Step 5 — check. The answer must be less than 930 MeV (one whole u) but far more than a few MeV, and 186 MeV sits correctly between -- a typical scale for a single nuclear event.

The distractors.

- **93 MeV** uses only 1 u of mass, ignoring the 0.20 factor.
- **372 MeV** doubles the defect to 0.40 u.
- **46 MeV** divides by the equivalent ($930 / 0.20$) the wrong way.
- **930 MeV** is the factor itself, the energy of one u, with the mass forgotten.

Relevant topics: mass-energy equivalence; $E = m c^2$; the $1 \text{ u} = 930 \text{ MeV}$ shortcut.

The trap. Using 1 u of mass (93 MeV) or doubling to 0.40 u (372 MeV), or returning the 930 MeV conversion factor itself. The released energy is the defect times the equivalent: $0.20 \times 930 = 186 \text{ MeV}$.

Q24 · The photoelectric effect: maximum kinetic energy of the electrons — **Answer D**

What is being tested. The photoelectric equation $K_{\text{max}} = hf - \phi$: the photon's energy pays the work function first, and whatever is left is the electron's kinetic energy.

Step 1 — the equation.

$$K_{\text{max}} = hf - \phi$$

Step 2 — substitute the given energies. The photon energy is $hf = 5.0 \text{ eV}$ and the work function is $\phi = 2.0 \text{ eV}$:

$$K_{\text{max}} = 5.0 - 2.0 = 3.0 \text{ eV}$$

Step 3 — the energy book-keeping. The photon arrives with 5.0 eV and the metal demands 2.0 eV to free the electron, so the electron leaves with the remainder -- a strict subtraction, not a shared or added amount:

$$K_{\text{max}} = E_{\text{photon}} - \phi = 5.0 - 2.0$$

Step 4 — the emission condition. Emission requires the photon energy to exceed the work function; here 5.0 eV > 2.0 eV, so the leftover 3.0 eV is genuinely available as kinetic energy, and any answer at or above 5.0 eV would break the energy budget.

So **Answer: D.**

Step 5 — check. The kinetic energy must be positive (emission happens) and below the photon energy (the work function was paid). 3.0 eV satisfies both.

The distractors.

- **2.0 eV** is the work function -- the energy that had to be spent, not what is left.
- **5.0 eV** is the photon energy, with no subtraction at all.
- **7.0 eV** adds the two, as if they reinforced.
- **1.0 eV** subtracts the wrong way, $\phi - hf$, giving a negative (which would mean no emission).

Relevant topics: the photoelectric effect; $K_{\text{max}} = hf - \phi$; work function.

The trap. Offering the photon energy (5.0 eV) or the work function (2.0 eV), or adding them (7.0 eV). The kinetic energy is the photon energy minus the work function: $K_{\text{max}} = 5.0 - 2.0 = 3.0 \text{ eV}$.

Q25 · A hydraulic jack: the load lifted by the larger piston — **Answer A**

What is being tested. Pascal's principle in a hydraulic jack: equal pressure in both pistons means the force is multiplied by the area ratio.

Step 1 — equal pressure.

$$F_{\text{small}} / A_{\text{small}} = F_{\text{large}} / A_{\text{large}}$$

Step 2 — solve for the large force. Rearranging the pressure balance gives the force on the large piston as the small force multiplied by the area ratio:

$$F_{\text{large}} = F_{\text{small}} \times (A_{\text{large}} / A_{\text{small}})$$

Step 3 — the area ratio. Because it is a ratio, the cm^2 units cancel:

$$A_{\text{large}} / A_{\text{small}} = 40 / 2.0 = 20$$

Step 4 — the load.

$$F_{\text{large}} = 100 \times 20 = 2000 \text{ N}$$

So **Answer: A.**

Step 5 — check. The jack multiplies the input force by 20, which is its purpose; the result is far larger than 100 N, as it must be, and no conversion was needed since both areas are in cm^2 .

The distractors.

- **100 N** is the applied force, ignoring the area multiplication entirely.
- **400 N** squares the gain -- using 20 and then multiplying by 20 again.
- **50 N** divides the input force by the small area, giving a pressure, not a force.
- **500 N** inverts the area ratio to $1/20$ and shrinks the force.

Relevant topics: Pascal's principle; hydraulic machines; force scales with area.

The trap. Giving the applied 100 N as the answer, or squaring the area ratio ($20 \times 20 = 400$ N), or inverting it. The force multiplies by the area ratio: $F_{\text{large}} = 100 \times (40/2.0) = 2000$ N, and the cm^2 units cancel in the ratio.